

EngiProof: From Published Research to Verified Engineering through Source-Bounded Evidence Automation

Living journal manuscript draft – methods first, results not final

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Draft status. This manuscript is an evidence-capture draft, not a submission-ready paper. Claims are deliberately bounded by the current EngiProof evidence state. Generalization, extraction-accuracy, and performance claims remain provisional until additional heterogeneous end-to-end cases are complete.

Abstract

Published engineering knowledge is commonly distributed across equations, tables, figures, narrative assumptions, and solver-specific results that are difficult to reuse computationally without losing provenance or overstating validation. This paper presents **ENGIPROOF**, a source-bounded engineering-evidence framework designed to convert selected published methods into reproducible calculations, independent checks, explicit comparisons, discrepancy records, evidence graphs, and callable engineering methods. The framework separates extraction candidates from engineering evidence, distinguishes **PUBLISHED**, **INDEPENDENT**, and **SOLVER_NEW** evidence, and keeps computational verification separate from engineering qualification.

The current development adds a conservative paper-ingestion pipeline comprising source fingerprinting, source-identity review, target discovery, missing-equation recovery, structure extraction, selected-target readiness gates, reproduction planning, discrepancy classification, human discrepancy decisions, and evidence-graph synchronization. Two pipe-in-pipe case studies are used as development stress tests. They demonstrate successful reproduction of selected published mechanics while also exposing source-backed numerical and unit inconsistencies that are retained rather than removed through parameter tuning. These examples motivate the central argument of the paper: for engineering reuse, paper-to-code conversion is insufficient; a useful system must explicitly track what was published, what was independently reconstructed, what agrees, what does not, and what remains unqualified.

This draft reports bounded case evidence only. Broader claims about generality and extraction performance are reserved for a later evaluation across additional heterogeneous papers.

1 Introduction

Engineering papers often contain enough information to reproduce part of a method, but not necessarily enough information to reproduce a complete model, resolve every convention, or qualify a result for design use. Reuse therefore requires more than extracting equations or generating executable code. A defensible workflow must preserve the identity of the source, distinguish

published reference data from independent calculations, expose assumptions and unavailable inputs, quantify comparisons, and retain unresolved discrepancies.

This distinction becomes important when a paper is internally inconsistent. A conventional paper-to-code workflow may silently select one source value, reinterpret a unit, or tune an input until the generated code reproduces the printed result. Such behavior can produce an apparently successful program while weakening engineering traceability. ENGIPROOF instead treats disagreement as evidence that must be classified and preserved.

The framework is motivated by the principle

published research $\rightarrow$ engineering evidence $\rightarrow$ callable engineering method,

not by paper-to-code conversion alone. The intended contribution is therefore an evidence progression model: what was published, what was independently reconstructed, what was newly generated, what was compared, what remains unresolved, and what has not been engineering-qualified.

The present manuscript focuses on the architecture and on two real-source development cases. The first is propagation buckling in subsea pipe-in-pipe (PIP) systems [1]. The second is global buckling and structural response of PIP systems [2]. The cases are intentionally used not only to demonstrate agreement, but also to test whether the framework can detect and preserve source-backed discrepancies.

The current bounded contributions are:

1. a source-bounded ingestion and provenance workflow;
2. explicit separation of extraction candidates, evidence classes, evidence status, and engineering qualification;
3. failure-aware equation and table recovery with selected-target readiness gates;
4. comparison and discrepancy records that preserve disagreement rather than tune it away;
5. append-only human discrepancy decisions and evidence-graph synchronization; and
6. development-case evidence showing both successful reproduction and source-backed mismatch detection.

2 Evidence model and framework architecture

2.1 Evidence classes

ENGIPROOF distinguishes the origin of a result from its maturity. Current evidence classes are:

- **PUBLISHED**: directly traceable to a published source or source-derived reference data;
- **INDEPENDENT**: a calculation or check implemented independently from the published result being tested; and
- **SOLVER_NEW**: newly generated solver evidence, used only where justified.

Evidence status is tracked separately using states including **DRAFT**, **SOURCE_READY**, **REPRODUCED**, **COMPARED**, **VERIFIED**, **CONDITIONAL**, and **BLOCKED**. Execution alone does not advance status. Similarly, verification does not grant engineering qualification.

2.2 Qualification boundary

Engineering qualification is intentionally orthogonal to numerical execution and evidence status. A study can be reproducible, independently checked, and compared while remaining `NOT_GRANTED` for engineering qualification. This prevents successful execution from being misrepresented as acceptance for project-specific design.

2.3 End-to-end evidence flow

Figure 1 summarizes the current workflow.

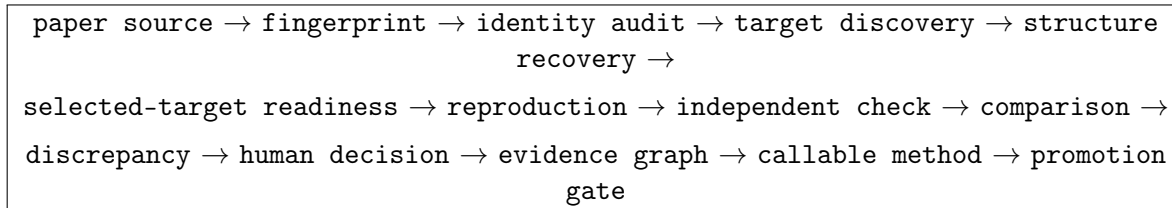


Figure 1: Current EngiProof evidence-automation workflow. Each arrow represents a controlled transition rather than an automatic promotion of evidence status.

A key invariant is that a candidate extracted from a PDF is not evidence merely because it has been recognized. Likewise, a callable method is not qualified engineering merely because it returns a number.

3 Paper ingestion and failure-aware structure recovery

3.1 Source fingerprint and identity

The ingestion layer records a SHA-256 content fingerprint and bibliographic metadata while avoiding persistence of machine-specific absolute paths. Copyrighted PDFs remain external by default. A source can be re-opened for enrichment only when its content fingerprint matches the original intake record.

A separate identity audit compares configured title, year, and DOI against information recoverable from the source. Importantly, a DOI that is not present in extractable PDF text is distinguished from a conflicting DOI. The second case study demonstrates this distinction: title and year were recoverable, while the DOI required external bibliographic confirmation rather than being silently inferred from the PDF text.

3.2 Target discovery and recovery

Figures, tables, equations, and appendices are initially lexical candidates. Real engineering PDFs frequently split equation bodies from their printed numbers, encode characters through Type1/CFF fonts, or refer to multiple equations in grouped text. The development process therefore added grouped cross-reference recovery, printed-equation-number recovery, and multiline equation-body reconstruction.

The first real automation case exposed the limitation of a simple lexical pass: only a small subset of relevant equations was initially represented. A later recovery pass substantially expanded the equation set for the same source. This before/after result is treated as a case-specific demonstration of failure-aware recovery, not as a general extraction-accuracy statistic.

3.3 Publisher-style table extraction

The second case study used ASME conference formatting and exposed a different generalization problem. Clear captions such as “Table 1 Pipeline data” did not contain the punctuation pattern expected by the earlier extractor. The subsequent implementation broadened caption recognition to punctuation-free and hyphenated publisher styles, rejected obvious cross-references, terminated table blocks at subsequent captions, and recognized parameter/unit/value rows such as diameter, thickness, area, and axial stiffness.

3.4 Selected-target readiness

For selected targets, structure generation and readiness are different states. A table is considered structurally ready only when a publisher caption and candidate data rows are available. An equation is structurally ready only when an equation-body candidate has been recovered. Readiness permits reproduction work to begin; it does not convert extracted text into verified evidence.

4 Development case studies

4.1 Case P40: propagation buckling in PIP systems

The P40 case is based on the propagation-buckling study of Karampour et al. [1]. The automated study selected a source-parameter table, a comparison table, and the main PIP propagation-pressure relationship. The analytical method was reproduced directly, while an independent work-balance reconstruction provided a second mechanics path.

For two of the three PIP configurations, the reproduced normalized propagation-pressure ratio agrees with the published Table 2 value to rounding. For the third case, the reproduced ratio is approximately 0.7057 while the published value is 0.66. An independent reconstruction gives approximately 0.7053, corroborating the reproduced value rather than the printed table value. The disagreement remains open and is not removed through parameter adjustment.

Table 1: P40 selected reproduction results. Values are retained as development evidence, not as a claim that the source publication is erroneous.

Case	Published Eq. (9) ratio	Reproduced ratio	Independent check
PIP-1	0.86	0.8635	consistent
PIP-2	0.69	0.6900	consistent
PIP-3	0.66	0.7057	≈ 0.7053

4.2 Case P41: axial load sharing and global buckling

The P41 case is based on Goplen et al. [2]. This source tests different mechanics and a different publisher format. The first evidence chain uses the published pipe geometry and Young’s moduli to independently reconstruct steel area and axial stiffness, then applies the published Eqs. (6)–(8) to the soil-friction load split.

The independently calculated steel areas and axial stiffnesses reproduce the published Table 2 values closely. For a published soil-friction increment of 471 N/m, the load-sharing equations give approximately 158.8 N/m for the inner pipe and 312.2 N/m for the outer pipe, closing to 471 N/m. The printed table instead reports 153 and 312 N/m.

A second P41 chain evaluates the published full-bonding friction criterion. With $\gamma_f = 1$ for the paper’s illustrative case classification, the required internal friction is approximately 158.78 N/m. This independently reproduces the four published Table 3 categories: no friction, partial bonding, full bonding, and full bonding. The paper’s dry-friction estimate is also independently checked: $1195 \times 0.3 = 358.5$ N/m, consistent with the rounded 358 N/m value used in the highest-friction case.

The final P41 checkpoint reviews the global-buckling response and Eq. (14). An elastic geometry check gives

$$\frac{(EI)_{\text{outer}}}{(EI)_{\text{inner}}} \approx 3.3966,$$

which independently supports the paper’s qualitative statement that the outer pipe is normally the stiffer bending member. This check does not reproduce the nonlinear moment–rotation curves or the finite-element model.

5 Source-backed discrepancy detection

One purpose of ENGIPROOF is to treat disagreement as engineering evidence rather than as an obstacle to be hidden. Table 2 summarizes the current source-backed examples.

Table 2: Current discrepancy examples retained by EngiProof.

Case	Published reference	Independent/reproduced evidence	EngiProof treatment
P40	Eq. (9) normalized ratio 0.66 for PIP-3	direct ≈ 0.7057 ; independent work-balance ≈ 0.7053	OPEN; published-reference mismatch; no tuning
P41-D001	Table 2: $f_S = 471$ N/m, $\Delta S_{\text{inner}} = 153$ N/m, $\Delta S_{\text{outer}} = 312$ N/m	published components sum to 465 N/m; Eqs. (6)–(8) reproduce $\approx 158.8 + 312.2 = 471$ N/m	OPEN; source-backed numerical mismatch; possible 153→159 typo remains unconfirmed
P41-D002	paragraph reports approximately 300/850 MNm	Figure 10 axis is labeled kNm with corresponding values on a 0–1000 scale	OPEN; source-internal unit mismatch; no silent unit correction

5.1 Why the distinction matters

For P41-D001, interpreting 153 N/m as a typographical 159 N/m would make the table sum consistent with the published total and is numerically close to the independent 158.8 N/m result. However, this remains an inference until supported by an erratum, author clarification, or other provenance. The framework therefore stores the printed value and the reproduced value separately.

Similarly, the P41 Figure 10 paragraph is naturally read as if kNm were intended, because the plot axis itself uses kNm. Nevertheless, the framework retains the paragraph’s printed MNm unit and records the mismatch instead of silently normalizing it.

These cases illustrate a broader methodological point: successful evidence automation should not be judged only by how often it reproduces a printed number. Its ability to expose irreconcilable or ambiguous source information is also an important engineering output.

6 Evaluation strategy

The final evaluation will be based on completed evidence chains rather than paper count. The current draft uses P40 and P41 as development cases, while additional heterogeneous cases are required before generality claims are made.

Planned evaluation dimensions include:

- source identity integrity and fingerprint matching;
- selected-target structural readiness;
- reproducibility of published numerical relationships;
- independent-check agreement;
- discrepancy detection and preservation;
- provenance/evidence-graph coverage;
- backward compatibility of existing studies; and
- ability to expose a callable method without overstating qualification.

No extraction success percentage is reported in this draft because the denominator and study-selection protocol are not yet fixed. A third mechanically different paper will be used next to stress-test the workflow beyond the two PIP examples.

7 Discussion

7.1 Paper-to-code is not paper-to-engineering

A generated function can be syntactically correct and numerically stable while still using a misread table cell, an ambiguous unit, or an unsupported convention. For engineering reuse, source provenance and discrepancy handling are therefore part of the computational method rather than documentation added after the fact.

7.2 Human review as a controlled state transition

Automation classifies and escalates discrepancies, but does not decide engineering acceptability. Human decisions are recorded append-only using dispositions such as resolved, bounded, accepted with rationale, or deferred. An unresolved high-priority discrepancy can block study promotion while leaving the original source evidence unchanged. These decisions do not grant engineering qualification.

7.3 Evidence graphs

The evidence graph links source identity, selected targets, callable methods, comparisons, discrepancies, automated assessments, and human decisions. Graph synchronization is designed to be additive and idempotent. Coverage is a provenance property only; a complete graph does not imply that the engineering method is verified or qualified.

7.4 Implications for research software

The source-backed mismatches in the two development cases suggest that an evidence-aware system may contribute value even when it cannot fully reproduce a publication. Detecting that a source is internally inconsistent, or that a result cannot be independently reconstructed without additional assumptions, can be more useful for engineering review than forcing an apparent numerical match.

8 Limitations

The current framework and manuscript have several important limitations:

1. PDF mathematical text can be corrupted by embedded fonts and extraction order.
2. Equation recovery can identify a source block without guaranteeing mathematically correct transcription.
3. Complex tables, merged headers, and figure axes remain harder than simple equation/table cases.
4. Symbol-definition extraction remains heuristic.
5. The framework does not infer genuinely missing source inputs; such cases remain conditional or blocked.
6. Independent agreement does not prove that a published interpretation is correct.
7. Reproduction and computational verification do not constitute engineering qualification.
8. The present automation evidence is still concentrated in structural/subsea engineering examples.
9. The P40 and P41 mismatch interpretations remain open unless supported by additional provenance such as errata or author clarification.
10. No global or local FE model is recreated for P41 Figures 8–10 in the present checkpoint.

9 Conclusions

This draft presents ENGIPROOF as a source-bounded framework for converting selected published engineering methods into reproducible calculations, independent checks, comparison records, discrepancy records, evidence graphs, and callable methods. The current evidence supports the architecture and demonstrates two important behaviors: the system can reproduce selected published mechanics, and it can preserve source-backed numerical and unit discrepancies instead of tuning them away.

The strongest current conclusion is therefore methodological rather than statistical: engineering evidence automation should make disagreement explicit and traceable. A paper-derived method should not be promoted merely because code executes or because a generated result can be made to match a printed number.

Before submission, the framework will be evaluated on additional heterogeneous papers, the claim–evidence matrix will be refreshed from the repository state, and the manuscript will be migrated to the selected journal template.

Data, software and source boundary

The EngiProof software repository is cited as software in this draft. Copyrighted source PDFs and publisher raster figures are intentionally not redistributed by EngiProof. Source fingerprints, bibliographic identifiers, bounded source locators, independently generated calculations, and machine-readable evidence records are used instead. The manuscript will be synchronized with the repository evidence graph before submission.

Acknowledgements

[TODO: Add acknowledgements only when finalized.]

References

- [1] Hassan Karampour et al. “Propagation Buckling in Subsea Pipe-in-Pipe Systems”. In: *Journal of Engineering Mechanics* 143.9 (2017), p. 04017113. DOI: [10.1061/\(ASCE\)EM.1943-7889.0001337](https://doi.org/10.1061/(ASCE)EM.1943-7889.0001337).
- [2] Stig Goplen et al. “Global Buckling of Pipe-in-Pipe: Structural Response and Design Criteria”. In: *Proceedings of the ASME 2011 30th International Conference on Ocean, Offshore and Arctic Engineering*. 2011, pp. 777–784. DOI: [10.1115/OMAE2011-49960](https://doi.org/10.1115/OMAE2011-49960).

A Claim–evidence matrix

ID	Candidate manuscript claim	Current state	Constraint
C01	Source fingerprinting and external-source provenance are implemented.	Supported	Architectural claim only.
C02	Extraction candidates are separated from engineering evidence.	Supported	May be stated directly.
C03	Execution does not automatically grant verification or qualification.	Supported	Core framework rule.
C04	Failure-aware recovery can recover structures missed by a simpler pass.	Case-supported	Do not generalize to all PDFs or report an accuracy percentage.
C05	Publisher-style table robustness improved after the ASME P41 stress test.	Case-supported	Describe the observed failure and fix; avoid broad benchmark claims.
C06	Independent checks can expose discrepancies rather than tune them away.	Supported by P40/P41	Keep case-specific and preserve open status.

ID	Candidate manuscript claim	Current state	Constraint
C07	The framework generalizes across arbitrary engineering literature.	Pending	Do not claim before more heterogeneous cases.
C08	EngiProof provides fully automatic paper-to-qualified-engineering conversion.	Do not claim	Contrary to the explicit human-review and qualification boundaries.

B Reproducibility checklist

Before submission, the following items must be regenerated directly from the tagged repository state:

- software version, commit, and release DOI if available;
- study registry and evidence-status table;
- P40 and P41 comparison CSV/JSON artifacts;
- discrepancy assessments and human-decision records;
- evidence-graph audits;
- exact test count and supported Python versions;
- all manuscript numerical tables from machine-readable results rather than manually copied values.